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A PROJECT REPORT ENTITLED  
AI-ASSISTED DIGITAL FORENSIC FRAMEWORK FOR AUTOMATED  
ANALYSIS OF MULTIPLE DIGITAL ARTEFACT

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## DECLARATION

We, Yeboah Franklin, Nyarkoh Bless Kwamina, Adongo Mendel, Umar Osman and Sankyi Elijah Kwabena declare that this work is our own work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other University.

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## **DEDICATION**

This project is dedicated to our valued family for their critical contribution to this success. Your support has been invaluable, and we are truly grateful.

## **ACKNOWLEDGEMENTS**

All glory to God Almighty, whose grace carried us through every step of this journey. We are deeply grateful to our supervisor, Dr Felix L. Aryeh, for his patience, honest guidance, and genuine investment in our success. His support made a real difference. To our lecturers and staff at UMaT, thank you for the knowledge and mentorship that shaped us into who we are the engineer we are today. We also extend our appreciation to Dr Hamidu Abdel-Fatao for his valuable contributions to this project. To our families, thank you for the love, prayers, and sacrifices you made without hesitation. And to ourselves Franklin, Bless, Mendel, Umar, and Elijah we started this together and we finished it together. That says it all.

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## LIST OF ABBREVIATIONS

| Abbreviation | Meaning                                                |
|--------------|--------------------------------------------------------|
| AUC-ROC      | Area Under the Receiver Operating Characteristic Curve |
| CSS          | Cascading Style Sheets                                 |
| DSR          | Design Science Research                                |
| FACE         | Forensic Artefact Correlation Engine                   |
| HTML         | HyperText Markup Language                              |
| IDE          | Integrated Development Environment                     |
| LOF          | Local Outlier Factor                                   |
| NLP          | Natural Language Processing                            |
| RAM          | Random Access Memory                                   |
| UTC          | Coordinated Universal Time                             |

# CHAPTER 1

## INTRODUCTION

### 1.1 Background

In the present day, the volume of data that needs to be examined in digital investigations, such as system logs, emails, web history, memory dumps and network traffic, is immense (Dunsin *et al.*, 2024). However, the problem is not the acquisition of the data, but the understanding of the same. Most of the tasks in the process of digital investigation are still manual, and investigators still rely heavily on multiple tools and personal experience (Baggili *et al.*, 2021). However, the process takes time, and the problem of delay is critical. Many forensic labs are still struggling with delay and cases can take a long time before being resolved (Al-Dhaqm *et al.*, 2020; Karie *et al.*, 2019).

Digital forensic labs still face serious delays because there is too much data to handle, which slows down investigations (Horsman, 2020). This problem has increased as the number of devices that can hold more data than ever before continues to increase. The investigators are facing the problem of information overload and the results obtained from the process are not consistent. Different tools interpret the same digital artefact differently. Many of the tools used do not work well together which makes the work harder for investigators (Khan *et al.*, 2022). This makes the correlation of the different pieces of evidence difficult. Also, the delay in the process can affect the outcome of the case, thereby affecting justice (Montasari *et al.*, 2020).

Although many frameworks are available for the process, most existing tools still focus on collecting data, but they do not really help in automatically finding patterns or linking evidence (Baggili *et al.*, 2021). Although the concept of artificial intelligence and machine learning has been introduced, the concept has not been implemented in the process of digital investigation (Khan *et al.*, 2022). There is no intelligent framework that can automatically analyse and correlate multiple pieces of digital evidence. Cybercrime is on the rise, and the situation demands the implementation of intelligent and smarter tools for the process. The digital forensics market is expected to reach 35.81 USD billion by 2032 (Soni, 2025).

## **1.2 Problem Statement**

There are several forensic tools available. However, investigators still face the problem of information overload, slow processing speeds, and difficulties in correlating evidence collected from different sources. There is a need for intelligent frameworks using AI tools to help investigators automate the process of evidence analysis and correlation. This project aims to bridge the gap by creating an AI-powered digital forensic framework that is capable of analysing heterogeneous digital artefacts.

## **1.3 Aims and Objectives**

The aim of this project is to develop an AI-assisted digital forensic framework with the ability to automate the analysis and correlation of multiple digital artefacts for faster and intelligent cybercrime investigations. Specific objectives are as follows;

- i. To design an AI-assisted digital forensic framework;
- ii. To develop and apply machine learning algorithms such as Isolation Forest and clustering techniques for identifying anomalies and detecting suspicious patterns in the digital forensic environment;
- iii. To develop mechanisms for correlating and prioritising digital evidence within the forensic environment;
- iv. To design and implement a web-based dashboard for the visualisation and management of the forensic environment; and
- v. To assess the efficiency of the proposed framework in comparison to conventional manual-based digital forensic techniques.

## **1.4 Methods Used**

### **1.4.1 Research Methodology**

The research methodology to be adopted in this project is Design Science Research (DSR). This methodology is appropriate because the study aims to design, develop, and evaluate an artefact that addresses a real-world problem in digital forensic investigations rather than merely describing the problem theoretically (Homem, 2018).

The study will begin with a review of modern forensic methods and AI-based tools to identify current challenges, research gaps, and relevant design requirements for the proposed framework (Al-Dhaqm *et al.*, 2020; Karie *et al.*, 2021). The findings from the review will be used to define the functional and non-functional requirements of the system.

Based on these requirements, a multi-layered system architecture will be designed with components for data acquisition, preprocessing, AI-based analysis, evidence correlation, storage, security and audit, and a web application layer for interaction and visualisation. The framework will then be implemented and tested using selected digital artefacts such as logs, browser history, and network traces.

Finally, the developed artefact will be evaluated by assessing its ability to analyse and correlate digital evidence efficiently and by comparing its performance with conventional manual-based digital forensic techniques.

## **1.5 Scope of the Study**

The framework will be designed to perform the following: the analysis of different artefact types such as logs, browser history, and network traces. It is not intended to replace human investigators but to assist them in their job in a better manner.

## **1.6 Tools and Facilities Used**

The following free and open-source tools for computations will be used to develop the proposed framework.

- i. Programming Languages and Core Development Tools: Python 3.x (primary programming language for backend, data processing, and machine learning), JavaScript/HTML/CSS (for frontend interface development), Visual Studio Code (IDE), and Git/GitHub (version control);
- ii. Machine Learning and Data Analysis: Scikit-learn for anomaly detection and clustering, Pandas for data manipulation, NumPy for numerical computations, and Matplotlib/Seaborn for visualization;

- iii. Web Framework and Interface: Flask for the frontend interface and API, Bootstrap for frontend UI development, Chart.js or Plotly.js for frontend timeline visualization, and DataTables for frontend search-based evidence tables;
- iv. Database and Storage: SQLite for lightweight relational database management, and SQLAlchemy for the ORM layer;
- v. Digital Forensic Utilities: Python-evtx for Windows Event Logs, Browser-history for multi-browser history extraction, Scapy for network packet analysis, and the Regex re module for pattern matching;
- vi. Testing and Documentation: Pytest for unit and integration tests, Jupyter Notebook for exploration and documentation, and Sphinx for technical documentation generation; and
- vii. Development Environment: Virtualenv/virtual environment for dependency isolation, pip for package management, and localhost for local development and testing.

## **1.7 Report Organization**

This report is organised into five chapters. Chapter 1 introduces the project, covering the background, problem statement, objectives, scope, tools, and expected outcomes. Chapter 2 reviews existing literature on digital forensics, the types of digital artefacts investigators work with, the shortcomings of traditional forensic methods, and how AI and machine learning are being applied to address these challenges. Chapter 3 covers the methodology behind the framework, including the system architecture, how different artefact data is ingested and preprocessed, and the machine learning models used for anomaly detection. Chapter 4 presents the implementation of the system, the web-based dashboard, the evidence correlation engine, and the results obtained from testing on forensic datasets. Chapter 5 wraps up the project with a summary of findings, a comparison against manual forensic methods, and recommendations for future work.

## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 Introduction

This chapter reviews the literature relevant to the development of an AI-assisted digital forensic framework. It discusses the foundations of digital forensics, the major types of digital artefacts examined in investigations, the limitations of traditional forensic approaches, and the application of artificial intelligence and machine learning to improve forensic analysis.

With the rapid growth of digital devices and internet-based activities, investigators now face increasing volumes of digital evidence that must be analysed accurately and within a reasonable time. As a result, researchers have shown growing interest in the use of artificial intelligence and machine learning to support and automate parts of the digital forensic investigation process.

The chapter therefore presents an overview of digital forensics, reviews key digital artefacts and existing challenges, and examines recent studies on intelligent forensic techniques that inform the proposed framework.

#### 2.2 Digital Forensics

The field of digital forensics has been defined as the scientifically appropriate and technically feasible process of identifying, collecting, preserving, analysing and presenting digital evidence in a manner that is legally acceptable (Montasari *et al.*, 2020). The field has been influenced by a number of prominent researchers, whose contributions have established both the theoretical foundation and practical application of digital forensics.

Digital evidence is described as data that is stored or transmitted in binary form and can be relied upon in legal proceedings (Casey, 2022). (Casey, 2022) established a conceptual vocabulary that remains widely used in digital forensics and emphasised the importance of rigour and repeatability in handling digital evidence.

Modern digital forensic frameworks have increasingly focused on automation and scalability

in digital investigations (Marturana *et al.*, 2020). In an influential paper on the future of digital forensics research, (Garfinkel, 2010) identified scalability as one of the most important challenges facing the discipline. (Garfinkel, 2010) argued that digital forensic tools were already struggling with the volume of data presented to investigators in the late 2000s and that the problem would become more severe as storage capacities continued to increase.

The process of digital forensic investigation can be divided into four phases. These are identification, where potential sources of digital evidence are identified; preservation, where digital evidence is preserved against alteration or destruction; analysis, where digital evidence is examined and inferences are made; and presentation, where findings are communicated to non-technical individuals such as judges and jurors (Montasari *et al.*, 2020). Of these phases, the one that takes the longest to complete and where the benefits of intelligent automation are greatest is the analysis phase. In a study of digital forensics challenges, more recent work also confirms that the analysis stage is still the most difficult part of digital investigations (Al-Dhaqm *et al.*, 2020).

### 2.2.1 Types of Digital Artefacts

A digital artefact is any data item created as a result of human activities on a digital device or network. Digital artefacts are the raw materials for digital forensic investigations. Knowledge of what digital artefacts are, their storage, and their relationship with each other is the foundation of digital forensics. This section briefly introduces the four types of digital artefacts that the proposed framework will cover.

### 2.2.2 System Logs

System logs are the automatically generated logs of activities on a computer or network device. These logs contain information regarding activities such as user login, file access, application usage, and system errors, along with a timestamp for each type of event. Horsman (2019) highlighted the practical value of system logs in digital forensic investigations and noted that the reality is that the vast majority of users are not going to think to meddle with system logs. However, the sheer amount of data contained in system logs has been a major practical hindrance, as a single server can generate gigabytes of system logs on a daily basis. Recent research (Ahmed *et al.*, 2023) confirms that machine learning methods are better at detecting

unusual patterns compared to rule-based systems. Their work demonstrated that machine learning techniques are particularly useful because, rather than relying on rules, they are able to "learn" what constitutes normal behavior, thus being able to detect new and previously unseen threats. This was extended specifically in the field of digital forensic investigation by (Dunsin *et al.*, 2024).

### 2.2.3 Browser History and Web Artefacts

Browser artefacts include the history of sites visited, cached files, cookies, saved password information, downloaded files, and search engine queries stored by various web browsers such as Chrome, Firefox, Edge, and Safari. These artefacts can prove to be very insightful regarding the activities that a suspect has been undertaking on the internet. Recent studies (Horsman, 2020) show that browser artefacts are still among the most common sources of evidence, but differences in browser structures make analysis difficult. Newer research shows that a lot of online evidence is now stored in cloud systems and social media, making investigations more complex (Baggili *et al.*, 2021). The automatic consumption and correlation of browser artefacts would directly address the issue of fragmentation as well as that of remote artefacts.

### 2.2.4 Network Traffic

Network traffic analysis involves examining the flow of data packets between devices connected to a computer network. Internet access, email transmission, and other forms of digital communication all generate network traffic. This traffic provides valuable information about communication patterns, the nature of transferred data, and possible suspicious activity. (Conti *et al.* 2018) noted that network data remains highly valuable in investigations, although it is often difficult to preserve because it is temporary and can be overwritten quickly. (Dunsin *et al.* 2024) reviewed several studies on the application of machine learning algorithms to network traffic analysis and reported that these methods achieved high accuracy in distinguishing legitimate from malicious activities. Nasir Memon and colleagues at New York University also argued that the next generation of digital forensic tools should be intelligence-driven rather than relying only on faster search mechanisms, identifying network traffic as one of the key areas requiring greater intelligence.

### 2.2.5 Memory Dumps

A memory dump refers to an exact copy of all the information contained in a computer's Random Access Memory (RAM) at a particular time. While information stored in a computer's hard drive is retained even after the computer has been switched off, memory is volatile and is lost when the computer is shut down. Memory contains information such as active processes, open network connections, plaintext encryption keys, and open documents, and hence it is an extremely valuable source of evidence. Memory forensics is still considered one of the most difficult areas because it requires deep technical understanding of how systems manage memory (Aljaedi *et al.*, 2020). Tools like the Volatility Framework are still widely used today, but interpreting the results still requires a lot of manual effort (Ligh *et al.*, 2019). The application of machine learning for automating memory analysis, such as recognizing known patterns of malware or recognizing unusual process behavior, is still an area of research (Dunsin *et al.*, 2024).

## 2.3 Limitations of Traditional Digital Forensic Methods

Though traditional digital forensic methods have worked satisfactorily for quite some time, their limitations have been exposed by the exponential increase in digital data. In this section, we will look at the three most important limitations of traditional digital forensic methods. These limitations include the limitations of manual investigations, limitations of existing tools, and limitations of the forensic backlog.

### 2.3.1 Manual Investigation and Human Limitations

In traditional digital forensic methods, an investigator simply goes through all the digital evidence and tries to detect any possible crime. This approach has worked satisfactorily for quite some time. However, with an increase in digital data, this approach has failed miserably. This is because human attention and concentration have limitations. In addition, different investigators may have different conclusions based on the same evidence. Recent research shows that human limitations still affect investigations, especially when dealing with large volumes of data (Al-Dhaqm *et al.*, 2020). Beebe and Clark (2005), who proposed a hierarchical framework for digital investigations, stated that lack of structured and standardized procedures was one of the most serious limitations of traditional digital forensic methods. More recent

studies also point out that lack of proper structure in investigations can lead to inconsistent results (Marturana *et al.*, 2020). The authors proved their point by demonstrating how unstructured procedures resulted in inconsistent results. The hierarchical framework proposed by Beebe and Clark was highly successful in addressing this limitation. However, it was still based on manual investigations. The implementation of an automated structured procedure eliminates all human limitations.

### 2.3.2 Existing Forensic Tools and Their Shortcomings

Popular digital forensic tools such as EnCase, Forensic Toolkit (FTK), Autopsy, and Sleuth Kit assist in digital evidence collection, keyword searching, file recovery, and timeline construction, among others. Recent studies (Duan *et al.*, 2022) show that many forensic tools still focus on single artefacts and do not support full integration across different data sources. This not only proves to be time-consuming but also creates room for potential errors in the chain of custody. A major drawback of all these tools is that they are signature or keyword-based, implying that they are only effective in finding things that the investigator has previously known to search for. They are incapable of finding anything else or making intelligent inferences on new patterns of evidence. Dunsin *et al.*, (2024), proposed that the next generation of digital forensic tools must go beyond search and replace it with machine intelligence that is capable of finding patterns that an investigator might not have known to search for in the first place.

### 2.3.3 The Digital Forensic Backlog

The most obvious effect of manual approaches on a large scale is the backlog in digital forensic examinations. More recent reports show that backlog is still a serious issue due to increasing digital evidence (Grobler *et al.*, 2020). Delays in digital forensic analysis still affect investigations and legal processes today (Behl and Behl, 2021). The proposed solution, data deduplication, was useful in reducing the backlog, it only solved part of the problem. The complete solution lies in intelligent automation in the analysis stage, which is the purpose of this project.

## 2.4 Artificial Intelligence and Machine Learning in Digital Forensics

Artificial intelligence is computer systems that perform tasks that normally require human intelligence. Machine learning is the subfield of artificial intelligence concerned with the development of methods for enabling machines to learn. This is a new set of tools for digital forensic experts, which can process data at rates and scales that are impossible for humans.

### 2.4.1 Anomaly Detection

Anomaly detection is the automated detection of data or patterns that differ significantly from what is considered normal. In digital forensics, anomalies can be highly significant, such as a login at an unusual hour, a large transfer to an unknown server, or a user account accessing files outside of its usual scope, which could all be signs of illegal activities or a security breach. Recent research confirms that machine learning methods such as Isolation Forest, One-Class SVM, k-Means Clustering, and Local Outlier Factor are better at detecting unusual patterns compared to rule-based systems (Ahmed *et al.*, 2023). In applying the results of Ahmed *et al.*, (2023) on anomaly detection, (Dunsin *et al.*, 2024) specifically focused on digital forensic data and showed that there is substantial empirical support for machine learning in anomaly detection in log file analysis, network traffic classification, and user behavior profiling. Widely used machine learning techniques for anomaly detection, including approaches applied in digital forensic data analysis, have been discussed in more recent studies on intelligent forensic frameworks and AI-assisted investigations (Ahmed *et al.*, 2023; Dunsin *et al.*, 2024).

### 2.4.2 Natural Language Processing

NLP enables computers to understand natural language. Recent studies show that NLP can help investigators quickly analyse large volumes of text data like emails and documents (Li *et al.*, 2021). Topic modeling helps investigators find relevant documents in a short time instead of reading each one individually. Named Entity Recognition automatically identifies names, dates, locations, and organizations. These tools are very helpful in fraud cases with tens of thousands of emails. Memon *et al.* argued that one of the most underutilized opportunities in the development of forensic software tools is natural language processing (Dunsin *et al.*, 2024).

### 2.4.3 Deep Learning

Deep learning involves the use of complex networks of interconnected nodes called neurons to recognize complex patterns in data. Convolutional Neural Networks were used to recognize images in image forensics to detect image tampering. Recurrent Neural Networks and Long Short-Term Memory networks were used to recognize sequential data such as logs. These networks are very helpful in analyzing logs where the sequence of events is important. Recent research shows that deep learning can help automate many forensic tasks, especially in pattern recognition (Khan *et al.*, 2022). However, deep learning models need massive training data, and such data is difficult to obtain as authentic data is sensitive. (Montasari *et al.*, 2020) argued that the “black box” nature of deep learning models is a concern as such models cannot explain the reasoning behind the conclusion.

## 2.5 Evidence Correlation and Prioritization

This section is about connecting the dots between digital evidence. In a complex investigation, no single piece of digital evidence will give you the complete picture. The suspicious email, the downloaded file that matches something, the suspicious login, and the outgoing connection to a known bad server these pieces of evidence may individually appear suspicious, but when correlated, give a clear picture of criminal activity. Manually correlating digital evidence is one of the most labor-intensive and mentally taxing processes in digital forensics. In the digital forensics arena, researchers were first to tackle this problem with the FACE system, demonstrating the technical feasibility of automated correlation of digital evidence and the potential to significantly reduce the workload of digital investigators. Recent work shows that graph-based models are now being used to connect different pieces of digital evidence more effectively (Xu *et al.*, 2020). Newer studies also highlight the importance of prioritising evidence using intelligent systems (Duan *et al.*, 2022). The process of ordering correlated digital evidence by relevance to the crime while this process is often ignored. The authors proposed the use of machine learning algorithms to predict the relevance of digital evidence based on past cases.

## **2.6 Digital Forensic Frameworks and Automation**

A digital forensic framework is a set of organized procedures and tools that guide the investigation from start to end. Modern frameworks now focus more on automation and intelligent processing of digital evidence (Marturana *et al.*, 2020). Research supports the use of layered automation to improve digital forensic processes (Behl and Behl, 2021), in which different levels of automation are used for different types of tasks, leaving the difficult decisions for humans and automating the easier tasks. This is very similar in structure to the digital forensic framework we are proposing. The importance of privacy and proper data handling in forensic systems has also been emphasised (Al-Dhaqm *et al.*, 2020). This is achieved in the proposed digital forensic framework by the focus on specific, rather than general, data artefacts.

## **2.7 Web-Based Dashboards for Forensic Visualization**

Even the most thorough analysis will become ineffective if the findings are not communicated properly. A web-based dashboard can help to bring together findings from various sources, emphasize key findings, and allow the investigator to search and navigate the findings without having to switch between tools. The introduction of visual tools help investigators understand complex data faster and make better decisions (Horsman, 2020). The proposed framework will make use of the Flask web development framework, a Python-based web development framework ideal for small applications and prototyping. Bootstrap will provide pre-made user interface elements of high quality and responsiveness. Chart.js and Plotly.js will provide the ability to create interactive web-based graphs and timelines. SQLite with SQLAlchemy will provide a database solution with minimal overhead, without the need to install a separate database server. The importance of usability is a key, though frequently ignored, element of a successful toolset. A powerful tool with poor usability will not be used correctly, leading to lower quality investigations. The proposed framework will place strong emphasis on usability as a first-class requirement, rather than an afterthought.

## **2.8 Review of Relevant Literature**

Several studies form the foundation of the proposed framework. Dunsin extitet al. (2024) examined the role of artificial intelligence and machine learning in digital forensics across

areas such as anomaly detection, log analysis, network profiling, and user behavior analysis. The study showed that AI techniques have been applied successfully in separate areas of digital forensics, but an integrated framework that brings together multiple artefact types, analysis methods, and correlation mechanisms remains limited. This gap directly supports the motivation for the proposed framework.

Al-Dhaqm *et al.* (2020) and Marturana *et al.* (2020) emphasised the importance of structured forensic procedures and investigation models that can cope with complex digital evidence. These studies show that digital forensic systems require not only technical tools but also well-defined processes for evidence collection, analysis, and interpretation. This provides a basis for designing the proposed framework in a way that combines automation with procedural structure.

Behl and Behl (2021) highlighted scalability and automation as major challenges in digital forensics, particularly as data volumes continue to increase. The study pointed out that conventional approaches are increasingly inadequate for handling the speed, scale, and complexity of present-day investigations. This supports the need for intelligent systems that can process large volumes of evidence more efficiently while still supporting forensic reliability.

Automated evidence correlation can be improved through advanced techniques such as graph-based models and machine learning, making it easier to connect related artefacts during an investigation (Xu *et al.* 2020). This contribution is closely related to the correlation component of the proposed framework, which aims to link evidence from heterogeneous sources in a meaningful way.

Ahmed *et al.* (2023) demonstrated that machine learning can be applied effectively to anomaly detection and evidence correlation within intelligent forensic frameworks. This provides support for the analytical component of the proposed system, especially in identifying suspicious patterns across logs, network data, and other artefact types.

Horsman (2020) emphasized that usability is an important factor in the practical adoption of forensic tools. A system with strong analytical capabilities but poor usability may not be used effectively by investigators. This observation informs the design of the web-based dashboard

in the proposed framework, where usability, accessibility, and clarity of presentation are treated as key design requirements.

Taken together, these studies provide a strong basis for the proposed framework by identifying the major advances in automation, anomaly detection, evidence correlation, and usability, while also revealing the continuing lack of a unified and intelligent forensic system that integrates these capabilities.

## **2.9 Summary of Related Works**

Table 2.1 illustrates a brief summary of the main related works discussed in this chapter.

**Table 2.1 Summary of Related Works on AI-Assisted Digital Forensic Frameworks**

| <b>Author(s)</b>               | <b>Methodology</b>                                   | <b>Contribution</b>                                             | <b>Identified Gap</b>                            | <b>How This Project Addresses It</b>                                 |
|--------------------------------|------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------|
| Dunsin <i>et al.</i> (2024)    | Review of AI and ML in digital forensics.            | Showed how AI is used in logs, network and behaviour analysis.  | No full system combining all artefacts.          | Builds one system that handles multiple artefacts together using ML. |
| Montasari <i>et al.</i> (2020) | Study on digital forensic frameworks and automation. | Highlighted need for intelligent and scalable forensic systems. | Limited real-world implementation of automation. | Implements a working automated framework with AI support.            |
| Sarker <i>et al.</i> (2020)    | Machine learning based cyber security analytics.     | Demonstrated ML effectiveness in detecting cyber threats.       | Focused more on security than forensics.         | Applies ML directly to forensic artefacts and investigations.        |
| Alazab <i>et al.</i> (2021)    | Deep learning for cybercrime detection.              | Improved detection accuracy using deep learning models.         | High complexity and low explainability.          | Uses simpler ML models that are easier to interpret.                 |

| <b>Author(s)</b>             | <b>Methodology</b>                                     | <b>Contribution</b>                                    | <b>Identified Gap</b>                  | <b>How This Project Addresses It</b>                                 |
|------------------------------|--------------------------------------------------------|--------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------|
| Baggili <i>et al.</i> (2021) | Research on digital forensic readiness and automation. | Emphasized proactive and intelligent forensic systems. | Lack of integrated analysis tools.     | Provides integrated ingestion and analysis of multiple data sources. |
| Karie <i>et al.</i> (2019)   | Evaluation of challenges in digital forensics.         | Identified data volume and analysis as key issues.     | Limited automation solutions proposed. | Focuses on automating analysis using AI techniques.                  |
| Hassan <i>et al.</i> (2022)  | AI-based anomaly detection in logs and networks.       | Showed ML improves detection of unusual patterns.      | Applied mostly in isolation.           | Combines anomaly detection across different artefacts.               |
| Ahmed <i>et al.</i> (2023)   | Intelligent forensic frameworks using ML.              | Proposed evidence correlation techniques.              | Limited real system implementation.    | Builds a functional system with correlation and visualization.       |

## 2.10 Challenges and Future Directions

Despite the positive developments discussed in this chapter, there are a few significant challenges that need to be addressed. To begin with, there is a scarcity of labeled data that is used in training. This is a challenge in the power of machine learning since such data is sensitive and not easily sharable. Then there is the challenge of explainability. There are many powerful machine learning algorithms, especially deep learning algorithms, that are unable to explain their decisions. In the legal field, every conclusion must have an explanation. Thus, this is an important challenge that must be addressed (Dunsin *et al.*, 2024). In addition, it is important that evidence is preserved throughout the process. If there is any change in evidence that is not logged, it becomes inadmissible. Because technology is changing so fast, there is a need to ensure that the tools are flexible and adaptable right from the start. Recent research shows that rapid changes in technology make it difficult for forensic tools to keep up (Behl and Behl, 2021).

## **CHAPTER 3**

### **SYSTEM DESIGN AND METHODOLOGY**

The current chapter discusses the methodology used in developing the AI-assisted digital forensic framework that analyzes multiple digital artifacts. In this chapter, the researcher outlines the research design, architecture of the system, the methods used to collect and process data, the algorithms used, and the methodology used to evaluate the digital forensic framework. This chapter further discusses the tools and technology used in the process of development, highlighting the manner in which all the objectives outlined in chapter one were addressed.

#### **3.1 Research Design**

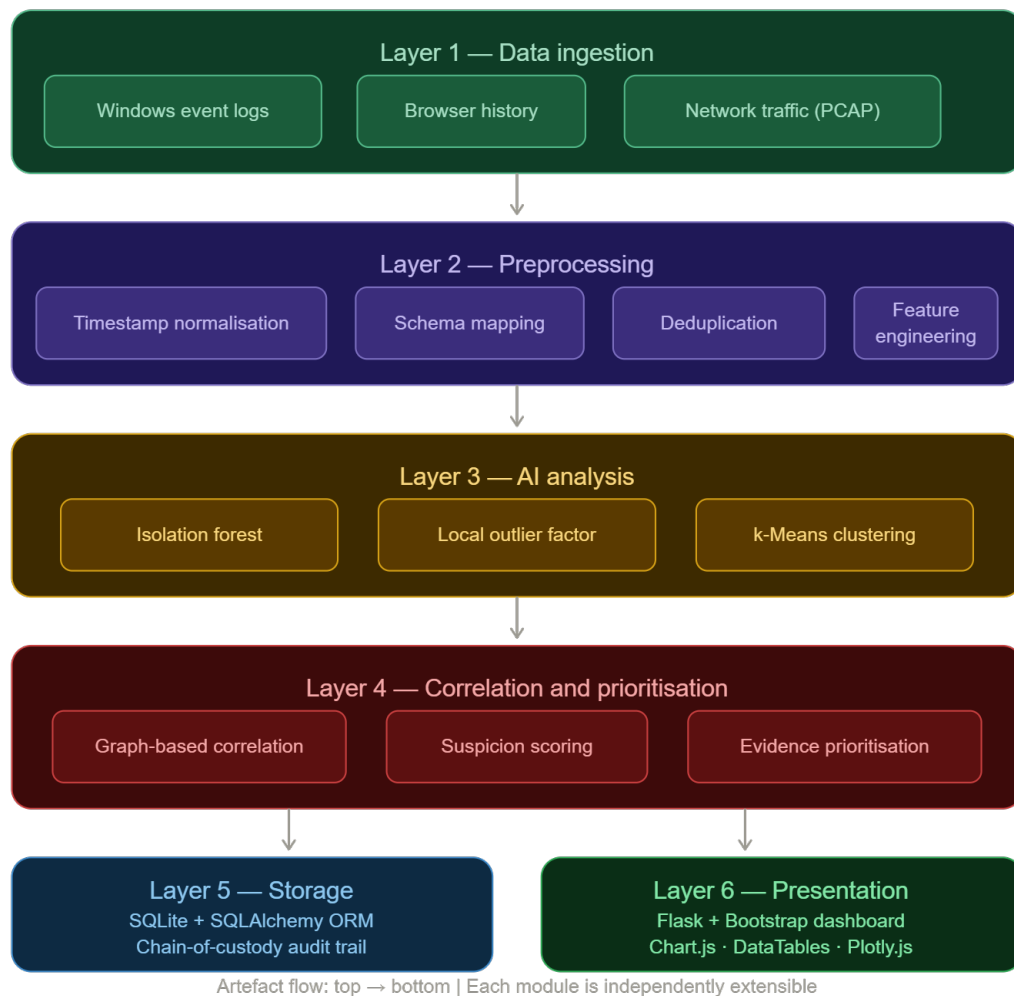
Design Science Research (DSR) represents the methodology adopted for the development of this research project. DSR is especially applicable in projects aiming at developing an artefact aimed at solving a real-world problem rather than providing explanations of a certain phenomenon (Homem, 2018). The practical problem associated with information overload, low processing speeds, and difficulties in evidence correlation in digital forensics serves as an example of the real-life issue that can be addressed through the development of an artefact, namely the framework suggested by the researcher. Therefore, DSR represents the most suitable methodology for conducting this study. According to the DSR methodology, the development process involves the completion of five phases. The problem and motivation phase entailed an exhaustive review of the existing literature and forensic software solutions, which helped define the drawbacks associated with the use of these methods and formulate the requirements for the framework development. It should be noted that this phase is covered in Chapters 1 and 2 of the paper. In addition, in the objectives phase, researchers translated the problem into a number of system objectives described in Chapter 1 (Section 1.3.2). Then, the architectural design and implementation of the framework took place according to the tools mentioned in Chapter 1 (Section 1.4). Finally, the demonstration phase was associated with the execution of the framework on the forensic datasets, while the evaluation phase entailed assessing the framework performance in comparison with traditional techniques.

This iterative, artefact-centred methodology ensures that every design decision is traceable to a documented requirement, and that the final evaluation is grounded in empirical evidence rather than theoretical assertion.

### 3.2 System Architecture

The new approach makes use of a multi-layered and modular architecture design, which takes into account the logical stages involved in a typical forensic examination. There are six layers involved in this architectural design, namely; the Data Ingestion Layer, the Preprocessing Layer, the AI Analysis Layer, the Correlation and Prioritization Layer, the Storage Layer, and finally, the Presentation Layer. These separate functionalities make it easy to scale and adapt the individual layers independently as needed.

Figure 3.1 provides a diagram of the complete framework architecture.



**Figure 3.1 Layered Architecture of the Framework**

### 3.2.1 Data Ingestion Layer

In addition, the Data Ingestion Layer deals with the process of gathering, validating, and importing the raw digital artefacts into the proposed architecture. In order to guarantee a high degree of efficiency and easy processing and scalability, priority will be given to the artefacts in structured and semi-structured formats, which can be efficiently processed by AI systems. For this particular case, there are three main categories of artefacts selected, system logs data, network traffic artefacts and browser history data. The first one is obtained via the HDFS Log dataset that represents a highly structured and voluminous set of artefacts for anomaly detection purposes. Network traffic data is taken from the UNSW-NB15, the popular benchmark data with preprocessed flow features. Finally, browser artefacts are created in the laboratory conditions that mimic regular and suspicious user behavior. It is proposed to develop three ingestion modules based on Python for each category. These modules will convert the input data from their initial form to a single tabular format for further usage. The structured artefacts, including HDFS and UNSW-NB15 will be ingested with Pandas, while browser data will be processed with the help of browser-history library.

### 3.2.2 Preprocessing Layer

The Preprocessing Layer ensures that the data files or evidences gathered from various sources are standardized to a uniform format or structure for organizing the data. The importance of this layer lies in the fact that one of the major barriers to cross-artefact analysis is the variability in artifact formats. The preprocessing steps involve the normalization of timestamps to a common UTC format, extraction and mapping to the internal schema, elimination of duplicates, and feature engineering to derive new features like session time, request rate, and process hierarchy. Pandas is used for tabular data manipulation and NumPy for numerical transformations.

### 3.2.3 AI Analysis Layer

AI Analysis Layer makes use of various machine learning algorithms on the preprocessed data to identify anomalies and suspicious behavioral patterns. The three algorithms used from the scikit-learn library follow theoretical underpinning. The Isolation Forest algorithm is used to detect anomalies on the basis of logs and network information in an unsupervised way

based on the assumption that isolating observations is easier when they are anomalous. The second algorithm that complements Isolation Forest on the density-based method is the Local Outlier Factor algorithm, which is very useful in identifying point anomalies. Lastly, k-Means Clustering algorithm segments behaviors by creating clusters and observing behaviors that form smaller or distant clusters.

#### 3.2.4 Correlation and Prioritization Layer

**Correlation and Prioritization Layer:** The Correlation and Prioritization Layer relates suspicious events associated with different artefacts to provide a comprehensive view of the process. In the FACE system, the Correlation and Prioritization Layer uses a graph-based model to correlate suspicious events. Suspicious events are modeled as nodes in a directed graph. Edges are established between suspicious events sharing common features like username, IP addresses, file paths, and timestamps within a defined time window. The connected components approach is employed to identify clusters of nodes. Suspicious event clusters are ranked based on their aggregated priority, which is determined by the number and nature of suspicious events in a cluster.

#### 3.2.5 Storage Layer

All artifacts, processed inputs, machine learning outputs, and correlation plots are stored in a SQLite database using the SQLAlchemy Object-Relational Mapper library. This database is structured in such a way that all activities related to the input data are logged so that there is a complete chain of custody of the data in compliance with the law. All records in the database have a unique identifier, time stamp of creation, and a link to the original artifact where it comes from, so that any output presented to a detective can always be tracked down to its original input artifact.

#### 3.2.6 Presentation Layer

The Presentation Layer gives an interface that allows the investigator to interact with the system. It consists of a web application whose backend API is provided by the Flask framework while its frontend is provided using Bootstrap. Four main views are provided by the dashboard, namely the Case Summary View, which contains information on the number of artifacts ingested, number of anomalies identified, and number of evidence clusters; Evidence

Timeline View, implemented using Chart.js, where the timeline of suspicious activity is plotted; Evidence Table View, implemented using Data Tables, where individual pieces of evidence can be filtered based on various parameters; and Cluster Detail View, where the graph representing the evidence that correlate in the investigation cluster is provided.

### **3.3 Data Collection and Artefact Sources**

Since the proposed model is meant to be used by the investigator, not as a model trained on proprietary datasets, all datasets used to evaluate our models come either from publicly available forensic datasets, or from synthetically created datasets within laboratory conditions, thus bypassing legal and ethical concerns regarding the use of actual case data. The dataset utilized for system log analysis comes from the HDFS log dataset made available by the LogHub repository, and represents a sizable and labelled corpus of system log entries, featuring both normal and abnormal activity. For browser history, the controlled dataset consisted of simulated investigations conducted through a virtual machine operating Windows 10, which involved regular browsing, download of certain file types marked as suspicious, as well as access to suspicious websites.

Network traffic was evaluated through the UNSW-NB15 dataset, which constitutes one of the most frequently cited network intrusion detection benchmark datasets and features both benign and malicious network flows. Lastly, memory analysis was performed using publicly available memory images from the MemLabs educational challenge series, providing memory artefacts without compromising any ongoing investigation. Prior to using the above-mentioned datasets in order to evaluate our framework, each dataset was processed through the same preprocessing pipeline.

### **3.4 Feature Engineering**

Prior to the application of machine learning algorithms, a feature engineering process is conducted to transform raw artefact data into structured numerical representations suitable for analysis. The extracted features vary depending on the artefact type, as summarised in Table 3.1.

**Table 3.1 Feature Extraction by Artefact Type**

| Artefact Type    | Features Extracted                                                                           |
|------------------|----------------------------------------------------------------------------------------------|
| System Logs      | Event ID, timestamp, user account, process ID, hour of day, anomaly flag (if available)      |
| Network Traffic  | Source IP, destination IP, destination port, protocol type, packet size, connection duration |
| Browser History  | Domain category, visit frequency, session duration, time between visits, hour of access      |
| Memory Artefacts | Process count per parent, number of network connections, suspicious process patterns         |

Categorical variables such as protocol type and domain category are encoded using numerical representations to ensure compatibility with machine learning models. Timestamp data is transformed into derived features such as hour-of-day to capture temporal patterns in user behaviour. This structured feature representation enables the effective application of unsupervised learning algorithms and ensures consistency across different artefact types.

#### 3.4.1 Model Training and Threshold Setting

All selected algorithms Isolation Forest, Local Outlier Factor (LOF), and k-Means Clustering are unsupervised learning techniques that do not require labelled training data. This makes them particularly suitable for digital forensic investigations, where labelled malicious data is often limited or unavailable. Isolation Forest is selected due to its efficiency in handling high-dimensional datasets and its ability to isolate anomalies based on random partitioning. Local Outlier Factor complements this by detecting anomalies based on local density deviations, making it effective for identifying subtle behavioural irregularities. k-Means Clustering is used to group similar behavioural patterns, allowing outliers to be identified as points that do not belong to any dominant cluster. Each model is trained using baseline data representing normal system behaviour.

The contamination parameter for Isolation Forest and LOF is initially set to 0.05, indicating that approximately 5 percent of the dataset is expected to be anomalous. This parameter is configurable within the system interface to accommodate different investigative scenarios. For k-Means Clustering, the optimal number of clusters is determined using the elbow method, which analyses the within-cluster sum of squares to identify the point of diminishing returns.

### 3.4.2 Suspicion Scoring

Each machine learning model produces an anomaly score for every record. To ensure comparability across models, these scores are normalised to a common scale using min-max normalisation as shown in equation 3.1.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (3.1)$$

Where:

- $X$  is the original score
- $X'$  is the normalised score

The final suspicion score for each record is computed using a weighted aggregation of the normalised scores shown in equation 3.2.

$$S = \sum_{i=1}^n w_i \cdot s_i \quad (3.2)$$

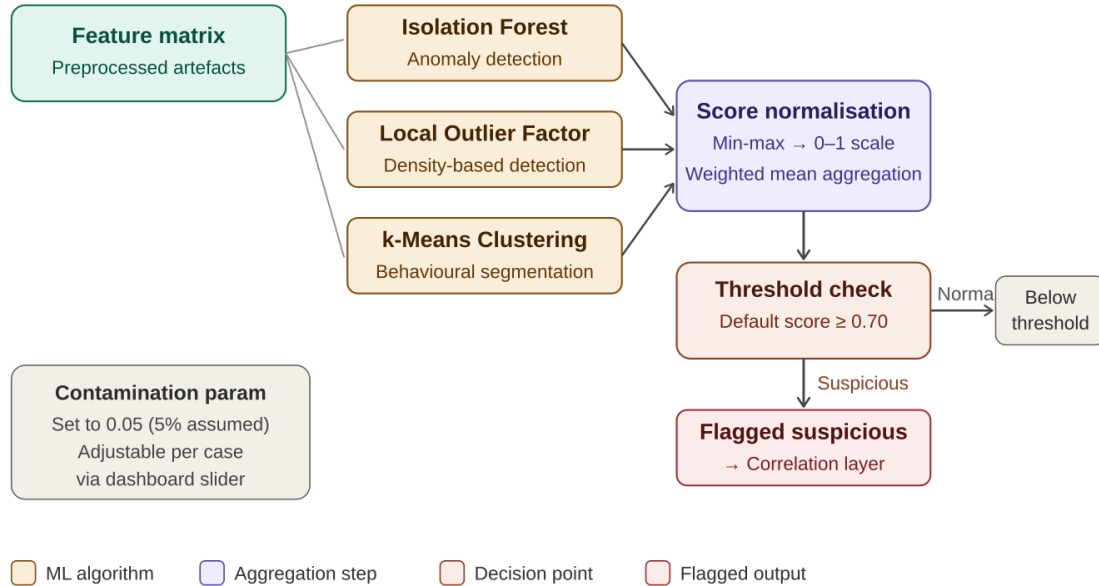
Where:

- $S$  is the final suspicion score
- $w_i$  is the weight assigned to model  $i$
- $s_i$  is the normalised score from model  $i$

Weights are assigned based on the relative performance of each model for the specific artefact type, as determined during evaluation.

Records with a final suspicion score above a configurable threshold (default = 0.70) are flagged as suspicious and forwarded to the Correlation and Prioritisation Layer.

Figure 3.2 illustrates the machine learning analysis pipeline and suspicion scoring workflow.



**Figure 3.2 ML Analysis and Suspicion Scoring Workflow**

### 3.5 System Implementation

#### 3.5.1 Development Environment

The framework was developed on a local development machine running Ubuntu 22.04 LTS. Python 3.11 was used as the primary programming language for the backend, data processing, and machine learning components. A Python virtual environment was maintained using virtualenv to isolate project dependencies and ensure reproducibility. Version control was managed using Git with a remote repository hosted on GitHub. All backend dependencies were managed using pip and documented in a requirements.txt file. The frontend was developed using HTML5, CSS3, and JavaScript, with Bootstrap 5.3 providing the responsive layout framework. The Flask development server was used for local testing, with the application served on localhost port 5000. The SQLite database file was stored in the project directory and managed exclusively through SQLAlchemy to prevent direct manipulation of the database outside the application layer.

### 3.5.2 Implementation of the Ingestion Modules

Each ingestion module was implemented as a Python class with a standardised interface comprising two methods: a parse method that accepts a file path and returns a Pandas DataFrame in the internal schema, and a validate method that checks the integrity of the source file before parsing. This standardised interface means that new ingestion modules can be added by any developer who conforms to the interface specification, without needing to understand the rest of the codebase. The EVTX parser uses the python-evtx library to extract event records and maps the relevant fields to the internal schema. The browser history parser uses the browser-history library to extract history records from Chrome, Firefox, Edge, and Safari and normalises the timestamps and URL fields. The network parser uses Scapy to read PCAP files packet by packet, reconstructing flows and computing per-flow statistics. The memory parser invokes the Volatility Framework via a subprocess call and parses the JSON output into the internal schema.

### 3.5.3 Implementation of the Web Dashboard

The Flask application was structured using the application factory pattern, with separate blueprints for the ingestion API, the analysis API, and the frontend routes. The frontend dashboard is a single-page application that communicates with the Flask backend through a RESTful API, using JavaScript Fetch API calls to retrieve data asynchronously without full page reloads. The evidence timeline is implemented using Chart.js, rendering an interactive bar chart of suspicious event counts grouped by hour of day. The evidence table is implemented using DataTables, providing client-side search, sort, and pagination over the full set of flagged records. The correlation graph is rendered using Plotly.js, displaying the connected-components graph as an interactive network diagram in which nodes can be clicked to reveal the underlying evidence record.

## 3.6 Ethical Considerations

This project adheres to established ethical principles in digital forensic research. All datasets used are either publicly available or generated within a controlled laboratory environment, ensuring that no real personal or sensitive data is exposed.

Furthermore, the framework is designed to operate only on data that has been lawfully acquired and authorised for forensic analysis. In the context of Ghana, this aligns with the principles outlined in the Data Protection Act, 2012 (Act 843), which governs the processing and handling of personal data. To ensure evidential integrity, the system maintains a complete audit trail of all data processing steps, supporting chain-of-custody requirements for potential legal proceedings.

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